Accessory Belt Drive System
Summary of Results

User: Abdurrahman Kol Design: AG00879_ANADOLU-ISUZU_6x6_Truck_Secondary_ALT_Drive_8PK1392HD_Gates-T38665-31Nm_17May2023

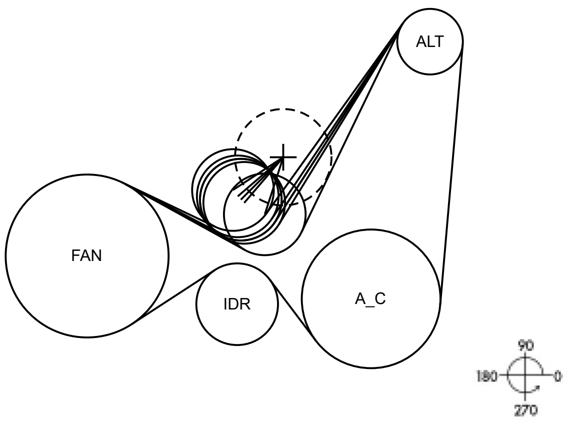
ANADOLU ISUZU 6x6 Cummins
AG00879 Secondary ALT&AC Drive
Tensioner Gates T38665; 31Nm
Gates Fleetrunner 8PK1392HD
17May2023

Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016

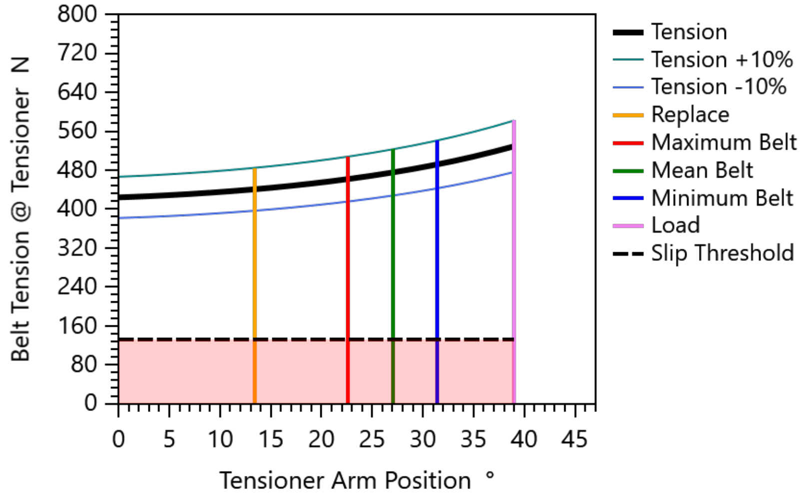
Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Effective Belt Length mm (ISO 9981)	1392

Tensioner Data

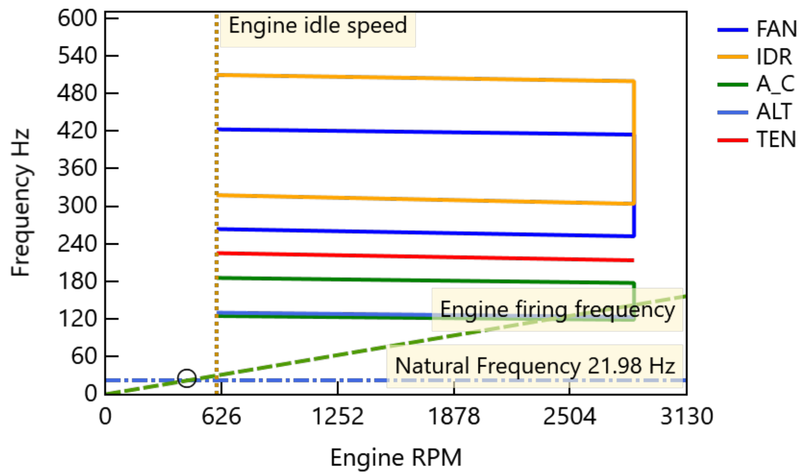
Type	Automatic
Design Tension N	476 (107 lbf)



Belt Tension Control



Natural Frequency Map



Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr	PK-2_2p-MT3 hours
95	1	5632

Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °
FAN	800	1000	1616	1890.7	20 / 241
IDR	800	1000	1615	2204.0	260 / 86
A_C	800	1000	1305	2728.4	108 / 138
ALT	800	1000	479	1743.0	258 / 152
TEN	600	1000	476	647.6	105 / 85

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

Permissible Grooved Pulley Axial Offset Allowance
For 0.90° Belt Entry Angle

Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
A_C->FAN	IDR (0.33)	1.63	62	38
ALT->A_C		3.83	100	
FAN->ALT	TEN (1.20)	2.32	23	77

Accessory Belt Drive System Geometric Analysis, Sheet 1 of 2

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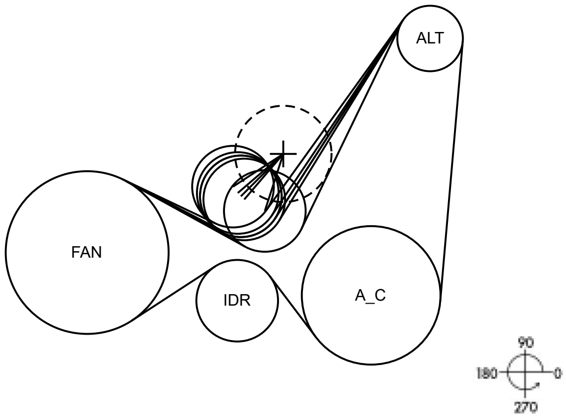
Layout Data mm

	X	Y	Flat	Pitch	Effective	DOB
FAN	0.00	0.00		152.01	149.61	150.60
IDR	140.00	-45.00	74.00	76.20	78.60	
A_C	265.00	-40.00		129.40	127.00	127.99
ALT	320.00	200.00		61.20	58.80	59.79
TEN	143.40	52.55	74.00	76.20	78.60	



Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Stretch and Wear Allowance % of Length	0.70
Belt Length Tolerance (+/-) mm	5.00

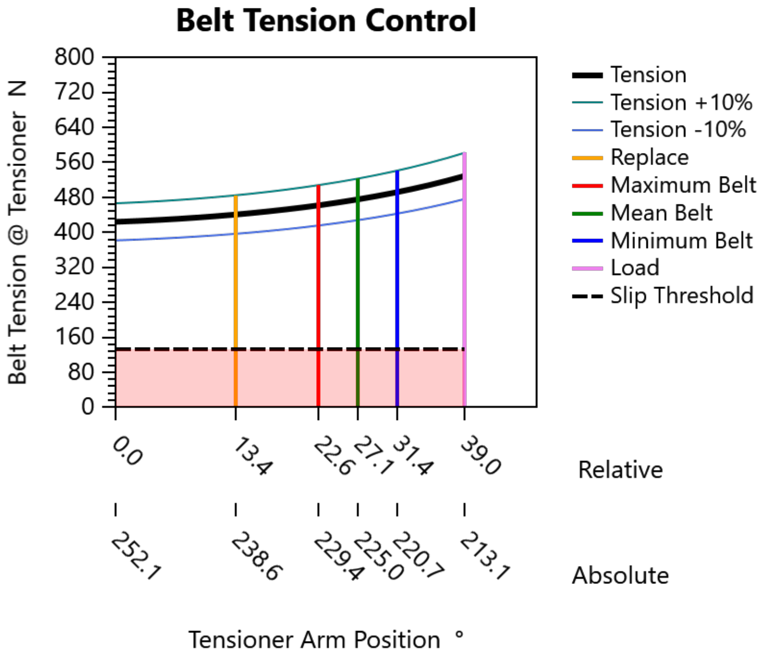
Tensioner Data

Type	Automatic
Design Tension N	476 (107 lbf)
Pivot Point {X, Y Coordinates} mm	183.00 , 92.15
Arm Length mm	56.0
Spring Mean Load Nm	31.14
Spring Pre-Load Nm	20.05
Spring Rate Nm/deg	0.409

RESULTS

Belt Drive System Geometry

	Span Length mm	Wrap Angle °	Speed Ratio (Ref. Engine)
FAN	92.8	241.3	1.000
IDR	71.3	86.0	1.995
A_C	243.8	138.0	1.175
ALT	219.6	152.2	2.484
TEN	101.5	85.4	1.995



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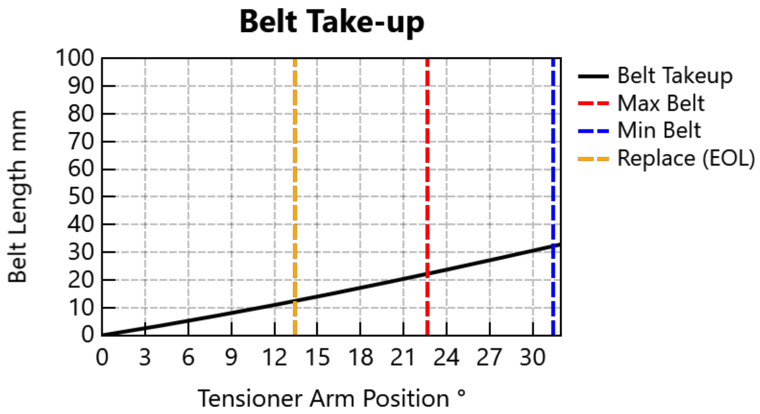
Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio mm/° 1.143

		Belt-Drive Length				
Position Description	**Free Arm**	**Replace**	**Max**	**Mean**	**Min**	**Load**
Arm Position °	252.1	238.6	229.4	225.0	220.7	213.1
Idler X Coordinate mm	165.8	153.9	146.6	143.4	140.5	136.1
Idler Y Coordinate mm	38.9	44.3	49.6	52.6	55.7	61.6
Hubload to Arm Angle °	35.6	47.0	55.3	59.5	63.7	71.3
Hubload Direction °	107.7	105.7	104.8	104.5	104.4	104.4
Hubload N	615.6	623.8	636.6	645.3	655.5	678.7
Tension N	424.1	440.8	461.8	475.5	491.8	529.1
Idler Wrap Angle °	93.1	90.1	87.2	85.4	83.6	79.8
Effective Drive Length mm	1419.9	1407.5	1397.7	1392.7	1387.7	1378.7
Required Eff. Belt Length mm	1419.2	1406.8	1397.0	1392.0	1387.0	1378.1



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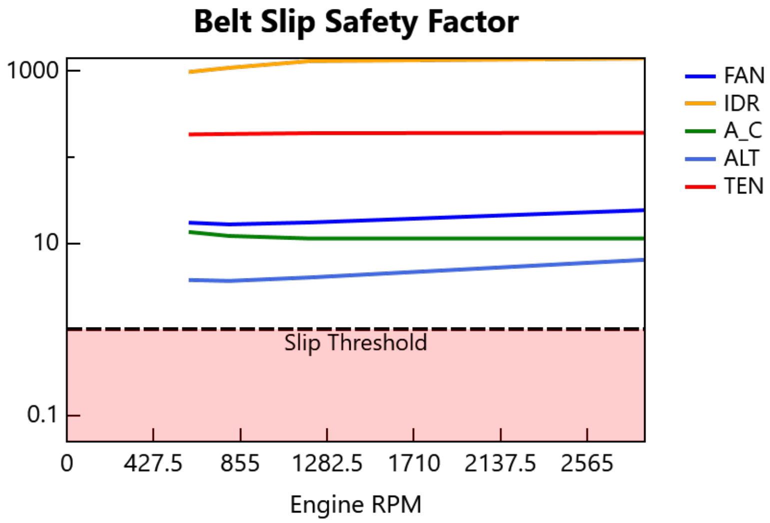
RESULTS

Belt Slip Sensitivity

Pulley	Most Critical Load Condition						Required Increase Tension or Wrap Angle	
	Acceler. RPM/s	Load Combination %						
		A_C	ALT				N	°
FAN	1000	100	100					
IDR	1000	10	10					
A_C	-1000	100	10					
ALT	1000	100	100					
TEN	1000	100	100					



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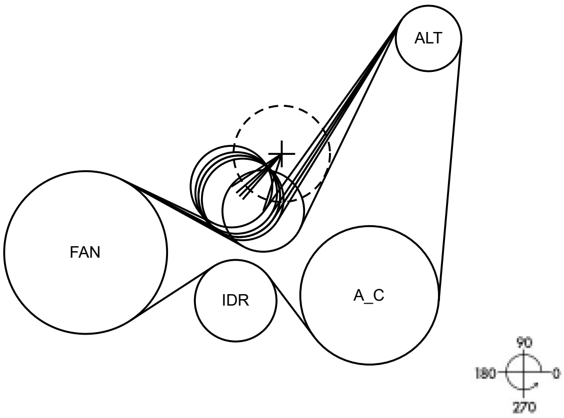

Load Conditions kW for DC 95%

%	Eng. RPM	V, km/hr	T, °C	FAN	IDR	A_C	ALT	TEN
5.00	600	1	80	4.92	0.01	1.40	3.50	0.01
35.00	800	1	80	6.72	0.01	1.90	4.80	0.01
35.00	1200	1	80	9.52	0.01	3.00	6.50	0.01
20.00	1700	1	80	12.42	0.01	4.20	8.20	0.01
5.00	2200	1	80	14.82	0.01	6.00	8.80	0.01

Mean Hubloads N for DC 95%

%	Eng RPM	FAN	IDR	A_C	ALT	TEN
5.00	600	1784	2053	2536	1648	647
35.00	800	1808	2088	2579	1668	646
35.00	1200	1751	2008	2457	1594	646
20.00	1700	1675	1901	2313	1520	646
5.00	2200	1605	1803	2151	1417	646

Mean Tensions N for DC 95%

%	Eng RPM	FAN	IDR	A_C	ALT	TEN
5.00	600	1506	1504	1211	478	476
35.00	800	1531	1529	1231	477	476
35.00	1200	1472	1471	1157	477	476
20.00	1700	1393	1393	1082	476	476
5.00	2200	1322	1321	979	476	476